

David Díaz-Villarejo
Ph.D. Candidate in Economics
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Education

Universidad Carlos III de Madrid Madrid, Spain
Ph.D. in Economics 2023 – present
Supervisors: Carlos Velasco Gómez, Jesús Gonzalo Muñoz

M.Res. in Economic Analysis 2023
B.Sc. in Economics 2021

Purdue University Indiana, U.S.
B.Sc. in Economics (Junior Year) 2019

Professional Experience

Universidad Carlos III de Madrid Madrid, Spain
Teaching Assistant (T.A.) 2022 – present

- Graduate Courses: Probability and Statistics Preliminary Course (Ph.D. in Economics; Fall, 2024-2026), Mathematics Preliminary Course (MSc. in Economic Development and Growth; Fall, 2024-2026).
- Undergraduate Courses: Econometrics (Spring, 2023-2026), Quantitative Microeconomics (Fall, 2025 & 2023), Time Series Econometrics (Fall, 2024), Microeconomic Theory (Fall, 2022), Mathematics for Economics II (Spring, 2022).
Teaching Excellence Awards in Econometrics and Quantitative Microeconomics.

Autoridad Independiente de Responsabilidad Fiscal (AIReF) Madrid, Spain
Research Assistant (R.A.) 10/2020 – 08/2021

- Long-term Forecasts and Sustainability Analysis of the Spanish Government Accounts for the Spanish Independent Fiscal Institution (IFI).

Analistas Financieros Internacionales, Afi (International Financial Analysts) Madrid, Spain
Professional Internship. 07/2019 – 12/2019

- Analysis of Spanish and European Credit Institutions for the Assets & Liabilities Committees (ALCOs).

Work in Progress

Gini Index Regression (*Job Market Paper*).
With Vedant Bhardwaj.

Comparisons of average outcomes do not reveal how inequality varies within populations with different characteristics. The paper develops Gini Index Regression (GIR), a framework in the spirit of mean or quantile regression that directly models the Gini of the conditional distribution of a nonnegative outcome, without choosing a parametric functional form for that distribution. In the linear case, the parameter of interest is shown to have a closed-form solution and, under model misspecification, the fitted function is a weighted best linear approximation to the conditional Gini. An omitted variable bias formula shows

that changing controls affects both the projection and the population relative to which outcomes are ranked. Estimation combines nonparametric series estimation of conditional mid-ranks with weighted least squares. The estimator reproduces empirical group Gini's in saturated group indicator specifications. We establish $\sqrt{n}$ -consistency, asymptotic normality, and validity of the bootstrap, accounting for both stages. A prediction characterization of the conditional Gini is also used to propose an R^2 -type goodness-of-fit measure. An application to U.S. wage data finds higher inequality among non-union workers, with a gap that widens with labor-market experience and varies by education level.

Locally Robust Semiparametric Estimation of Parametric Conditional Gini.

With Vedant Bhardwaj, Juan Carlos Escanciano, and Joel R. Terschuur.

Using the influence function for functionals of conditional distributions, we propose locally robust estimation and inference for Gini Index Regression (GIR) models, considering three alternative model specifications as leading examples, robust to first-stage, nonparametric kernel estimation of the conditional distribution function.

A Durbin-FGLS Approach to Robust Inference for Time Series Regression.

The paper proposes a new feasible generalized least squares estimator based on time series Durbin regressions. The estimator is shown to be consistent under much weaker exogeneity assumptions than the standard OLS exogeneity, allowing the number of first-stage lagged regressors to grow with the sample size. It is also consistent when the existing FGLS variants proposed in the literature are not, and when they are, also fully efficient. Theoretical results on consistency and robust inference are illustrated via Monte Carlo experiments.

Courses & Summer Schools

- Local Projection Methods for Time Series and Panel Data. CEMFI Summer School. Prof. Óscar Jordà (Federal Reserve Bank of San Francisco and UC Davis). 2024.
- DSGE and Time Series Models for Macroeconomic and Policy Analysis. CEMFI Summer School. Prof. Marco del Negro (Federal Reserve Bank of New York). 2024.
- Large Language Models and Machine Learning for Unstructured Data. Fundación Ramón Areces - IESE Business School. Prof. Stephen Hansen (UCL). 2024.
- Linear Programming and Economic Applications. Math+Econ+Code. Prof. Alfred Galichon (NYU). 2023.
- Applied Functional Analysis. Universidad Carlos III de Madrid. Prof. L. Alberto Ibort. 2023.

Honors & Awards

- Beca FPI para la Formación de Doctores. Ministerio de Ciencia e Innovación, 2023. Ph.D. Scholarship.
- Master's Scholarship. M.Res. in Economic Analysis (UC3M), 2021. Banco Santander. Full Scholarship.

Languages

Programming: R, Matlab, EViews, Stata, Excel.

Communication: English (C1), French (B2), Spanish (Native).

References

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